

Multiple Choice (20 marks)

1. What is one way to change water from a liquid to a solid?
 - (a) decrease the temperature
 - (b) increase the temperature
 - (c) decrease the mass
 - (d) increase the mass
2. Approximately how many movies are watched in homes in the United States in one year?
 - (a) 200,000,000
 - (b) 2,000,000,000
 - (c) 20,000,000,000
 - (d) 200,000,000,000
3. The view that moral considerations do not apply to war is known as
 - (a) jus ad bellum.
 - (b) antiwar pacifism.
 - (c) just war theory.
 - (d) moral nihilism.
4. An embryologist studying the development of a vertebrate organism from the zygote stage to fetus would be justified in drawing which of the following...
 - (a) Ontogeny recapitulates phylogeny.
 - (b) Early embryos display identical features of their class, order, and species.
 - (c) An early human embryo has features in common with early fish and avian embryos.
 - (d) A human embryo displays features of adult fish and birds in the course of its development.
5. Gastrulation is the process of
 - (a) mesoderm formation and occurs after neurulation.
 - (b) ectomesenchyme formation and occurs before neurulation.
 - (c) mesoderm formation and occurs before neurulation.
 - (d) ectomesenchyme formation and occurs after neurulation.

6. As number of organisms in a group increases , how will the chance of survival of each organism change?
- (a) float
 - (b) minimal change
 - (c) augmented
 - (d) lowered
7. Phosphocreatine in the muscle cell is found in:
- (a) the mitochondria.
 - (b) all subcellular compartments.
 - (c) the sarcolemma.
 - (d) the cytoplasm.
8. Every twenty-four hours and fifty minutes, the moon can have an effect on bodies of water causing them to
- (a) swell
 - (b) float
 - (c) die
 - (d) burn
9. An example of camouflage is when an organism looks like what?
- (a) clouds
 - (b) local flora
 - (c) buildings
 - (d) oceans
10. John's shadow looks like it's directly underneath him. What time is it?
- (a) 9am
 - (b) 12pm
 - (c) 12am
 - (d) 2pm

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a python function to check whether the value exists in a sequence or not.
2. Write a python function to find the minimum sum of absolute differences of two arrays.

End of Paper